Digital Egypt Pioneers Initiative
Round 4
2025 - 2026

رواد مصر الرقمية
الدورة الرابعة
2025 - 2026

Track: AI & Data Science - Microsoft Machine Learning Engineer

Group Code: ONL4_AIS2_S7

Instructor: Eman Elgalad

Company Name: Skills Dynamix

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Project Name

"AI-Powered Predictive Maintenance for Industrial Equipment"

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Planning

&

Management

Project Proposal:

- **Overview**
  - Predictive maintenance is an advanced maintenance strategy that uses data analysis, machine learning, and artificial intelligence techniques to predict equipment failures before they occur. Unlike reactive maintenance, which only addresses failures after they happen, and preventive maintenance, which depends on predefined maintenance schedules, predictive maintenance enables organizations to make proactive decisions based on the actual health condition of equipment.
  - This project aims to develop an Explainable Artificial Intelligence (XAI)-based Predictive Maintenance System capable of estimating the Remaining Useful Life (RUL) of industrial equipment using historical sensor measurements and operational conditions.
  - The system utilizes NASA's Commercial Modular Aero-Propulsion System Simulation (CMAPSS) dataset, which contains simulated aircraft engine degradation data collected from multiple sensors under different operating conditions.
  - The proposed solution integrates:
    - Machine Learning models for accurate RUL prediction.
    - Explainable AI techniques using SHAP to interpret model decisions.
    - A RAG-based AI maintenance assistant to provide knowledge-based support.
    - An interactive Streamlit dashboard for monitoring, visualization, and prediction.
    - Backend services for model inference and system integration.
  - The project provides a complete AI-driven maintenance platform that improves equipment reliability, reduces downtime, optimizes maintenance operations, and increases trust in AI predictions through transparency and explainability.

- **Problem Statement**
  - Unexpected equipment failures represent a major challenge in industrial environments, causing production interruptions, financial losses, increased maintenance expenses, reduced equipment lifetime, and potential safety risks.
  - Traditional maintenance strategies have several limitations:
    - Reactive Maintenance: Equipment is repaired only after failure occurs, resulting in unexpected downtime and high repair costs.
    - Preventive Maintenance: Maintenance is performed according to fixed schedules, which may lead to unnecessary servicing or failure before the scheduled maintenance time.
  - These approaches lack the ability to continuously evaluate equipment health and predict future failures based on real operational conditions.
  - Therefore, there is a need for an intelligent predictive maintenance solution that can analyze sensor data, estimate equipment degradation, predict Remaining Useful Life (RUL), and support maintenance teams in making accurate and timely decisions.

- **Objectives**
  - Main Objectives
    - The main objective of this project is to develop an Explainable AI-driven Predictive Maintenance System that can:
      - Predict equipment failures before they occur.
      - Estimate Remaining Useful Life (RUL) accurately.
      - Reduce unexpected equipment downtime and maintenance costs.
      - Improve maintenance planning and operational efficiency.
      - Provide transparent AI predictions using Explainable AI techniques.
      - Offer an interactive and user-friendly monitoring dashboard.
  - Technical Objectives
    - Dataset Understanding
      - Analyze the structure and characteristics of the NASA CMAPSS dataset.
      - Understand sensor measurements and operational conditions.
      - Study equipment degradation behavior over time.
    - Data Analysis
      - Perform Exploratory Data Analysis (EDA).
      - Identify patterns, trends, and anomalies in sensor readings.
      - Analyze sensor importance and relationships.
    - Feature Engineering
      - Generate RUL labels from engine lifecycle data.
      - Extract statistical and time-series features.
      - Select the most relevant features for prediction.
    - Machine Learning Development
      - Develop regression models for RUL estimation.
      - Compare different models using RMSE, MAE, and R^2 metrics.
      - Optimize model performance through hyperparameter tuning.
    - Explainable AI
      - Integrate SHAP for model interpretation.
      - Identify important factors affecting predictions.
      - Improve transparency and trust in AI decisions.
    - System Deployment
      - Develop backend services for prediction.
      - Build an interactive Streamlit dashboard.

- Integrate AI prediction, visualization, reporting, and maintenance assistance features.

- **Scope**
  - Included in Scope
    - NASA CMAPSS dataset collection and preparation.
    - Data cleaning and preprocessing.
    - Exploratory Data Analysis (EDA).
    - Feature engineering and feature selection.
    - Machine learning model development for RUL prediction.
    - Model optimization and performance evaluation.
    - Explainable AI implementation using SHAP.
    - Backend development for model inference.
    - Streamlit dashboard development.
    - RAG-based maintenance assistant integration.
    - Technical documentation and final presentation preparation.
  - Excluded from Scope
    - Real-time IoT sensor hardware integration.
    - Cloud deployment and enterprise production scaling.
    - Mobile application development.
    - Embedded system implementation.
    - Advanced deep learning research beyond project requirements.

- **Dataset Description**
    - Dataset Source
        - The project uses NASA’s Commercial Modular Aero-Propulsion System Simulation (CMAPSS) Turbofan Engine Degradation Dataset, a widely used benchmark dataset for predictive maintenance and Remaining Useful Life estimation.
        - The dataset contains multivariate time-series sensor measurements collected from simulated aircraft engines operating under different conditions and degradation scenarios.
    - Dataset Structure
        - The dataset consists of four subsets:
            - FD001 (Single operating condition with a single fault mode).
            - FD002 (Multiple operating conditions with a single fault mode).
            - FD003 (Single operating condition with multiple fault modes).
            - FD004 (Multiple operating conditions with multiple fault modes).
        - Training Data:
            - train_FD001.
            - train_FD002.
            - train_FD003.
            - train_FD004.
        - Testing Data:
            - test_FD001.
            - test_FD002.
            - test_FD003.
            - test_FD004.
        - Ground Truth Data
            - RUL_FD001.
            - RUL_FD002.
            - RUL_FD003.
            - RUL_FD004.
    - Dataset Features
        - Each record represents an engine condition at a specific operating cycle and contains:
            - Engine ID: Unique identifier for each engine.
            - Cycle: Represents operational time progression.
            - Operational Settings: Three parameters describing engine conditions.

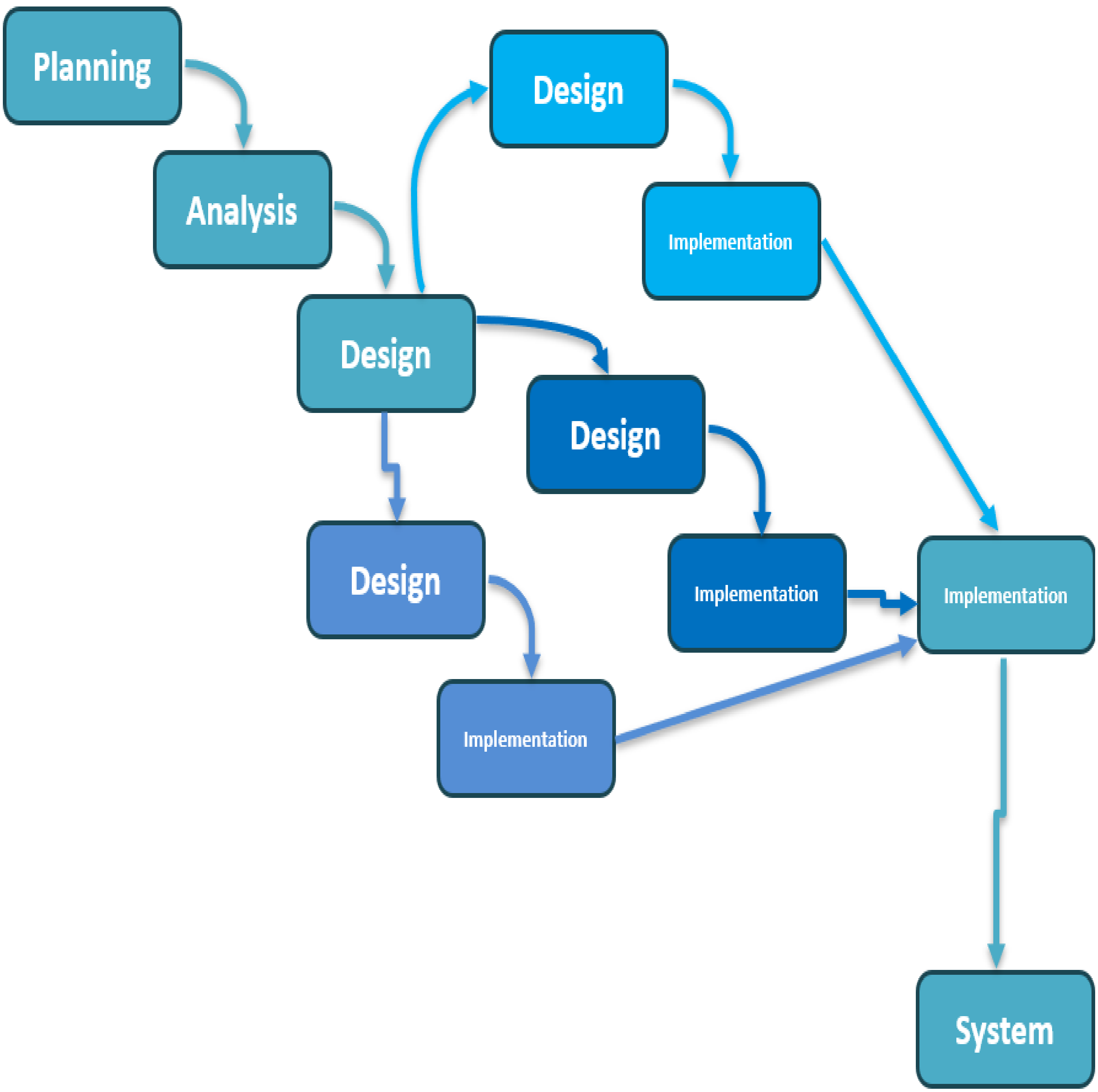
- Sensor Measurements: 21 sensor readings collected from engine components.
- Remaining Useful Life (RUL): Target variable representing the number of cycles remaining before failure.

- Dataset Characteristics
  - Type: Multivariate Time-Series Dataset.
  - Domain: Predictive Maintenance and Prognostics.
  - Application: Aircraft Engine RUL Prediction.
  - Operating Conditions: Single and multiple conditions.
  - Fault Modes: Different degradation scenarios.

- **Expected Outcomes**
  - Accurate RUL Prediction
    - Develop a reliable machine learning model capable of estimating equipment health and predicting future failures.
  - Explainable AI Framework
    - Provide transparent prediction explanations using SHAP, allowing users to understand the impact of different sensor features.
  - Intelligent Maintenance Support
    - Enable maintenance teams to make proactive decisions based on predicted degradation and failure risks.
  - Reduced Downtime and Costs
    - Minimize unexpected failures and optimize maintenance schedules.
  - Interactive AI Dashboard
    - Provide a Streamlit-based interface for:
      - Real-time RUL prediction.
      - Equipment monitoring.
      - Data visualization.
      - Maintenance management.
  - AI Maintenance Assistant
    - Provide a RAG-based assistant capable of answering maintenance questions using a domain-specific knowledge base.
  - Professional Documentation
    - Deliver complete technical documentation and final project presentation.

Project Plan:

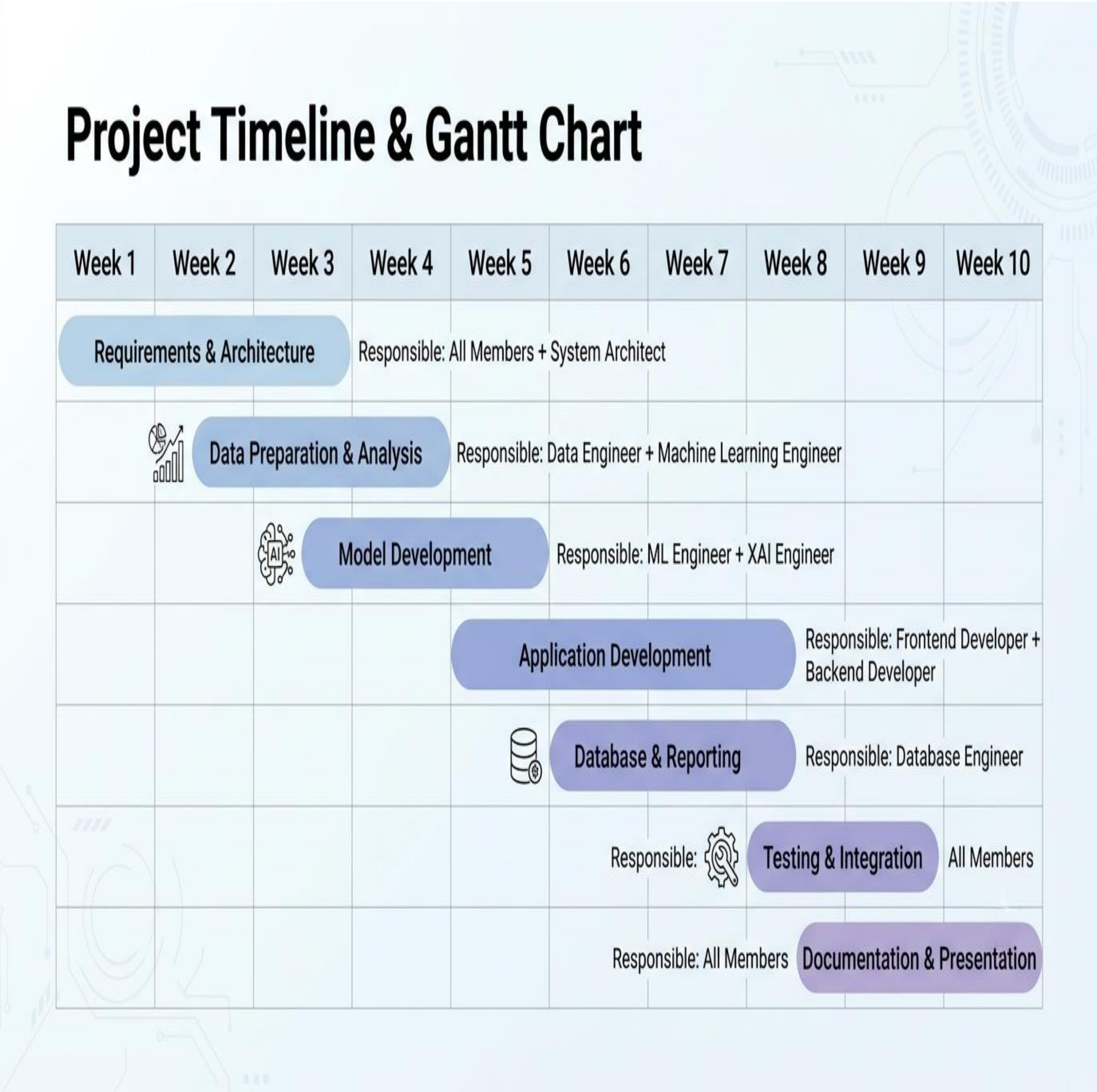
- **Methodology (Parallel Development)**



- **TimeLine**

Phase	Duration	Responsible Members
Requirements & Architecture	Week 1–2	Moaz Osama
Data Preparation & Analysis	Week 2–3	Abrar Ashraf
Model Development	Week 3–5	Abrar Ashraf & Nour-Eldin Kamel
Application Development	Week 5–8	Marwan Muhammad & Salma Muhammad
Database & Reporting	Week 6–8	Yussef Ahmad
Testing & Integration	Week 8–9	Moaz Osama & Salma Muhammad
Documentation & Presentation	Week 9–10	Moaz Osama, Abrar Ashraf, & Salma Muhammad

- **Gantt Chart**



Project Timeline & Gantt Chart
Week 1
Week 2
Week 3
Week 4
Week 5
Week 6
Week 7
Week 8
Week 9
Week 10
Requirements & Architecture
Responsible: All Members + System Architect
Data Preparation & Analysis
Responsible: Data Engineer + Machine Learning Engineer
AI
Model Development
Responsible: ML Engineer + XAI Engineer
Application Development
Responsible: Frontend Developer + Backend Developer
Database & Reporting
Responsible: Database Engineer
Responsible:
Testing & Integration
All Members
Responsible: All Members
Documentation & Presentation

- **Milestones**

No.	Milestone	Description
1	Dataset Preparation	Collect, clean, validate, and preprocess NASA CMAPSS data
2	Exploratory Data Analysis (EDA)	Analyze sensor behavior and degradation patterns
3	Feature Engineering	Generate RUL labels and extract useful features
4	Model Development	Train and evaluate ML models
5	Model Optimization	Tune parameters and improve performance
6	Application Development	Develop Streamlit dashboard and integrate system components
7	Final Documentation and Presentation	Prepare reports and final presentation

- **Deliverables**
    - Prepared and validated dataset.
    - Completed exploratory data analysis.
    - Engineered features for ML modeling.
    - Trained machine learning model.
    - Optimized and evaluated prediction model.
    - Explainable AI module using SHAP.
    - RAG maintenance assistant.
    - Completed Streamlit application.
    - Technical documentation.
    - Final presentation.

- ## Resource Allocation

Resource Category	Allocation	Description
Human Resources	Project Team	Responsible for development, testing, integration, and documentation
Dataset	NASA CMAPSS Dataset	Provides engine sensor data for RUL prediction
Programming Tools	Python Ecosystem	Pandas, NumPy, Scikit-learn, XGBoost, SHAP, FAISS
Development Environment	Jupyter Notebook / VS Code	Used for coding, analysis, and model training
Backend Services	FastAPI	Provides prediction APIs and system integration
Frontend	Streamlit	Interactive dashboard and visualization
AI Services	Groq LLM + RAG Pipeline	Knowledge-based maintenance assistant
Experiment Tracking	MLflow	Model tracking and lifecycle management
Computing Resources	Personal Computer	CPU-based development and training
Documentation Tools	Microsoft Word / PowerPoint	Reports and final presentation preparation

Task Assignment & Roles:

Team Member	Role	Responsibilities
Moaz Osama	Project Manager & System Architect	Architecture design, task coordination, integration, documentation, presentation
Abrar Ashraf	Machine Learning Engineer	Data preprocessing, feature engineering, model development, training, tuning, evaluation, XGBoost implementation
Nour-Eldin Kamel	Data Engineer & XAI Engineer	SHAP explanations
Salma Muhammad	Backend & Authentication Engineer	Backend services, authentication, APIs, system logic
Marwan Muhammad	Frontend & RAG Engineer	Streamlit dashboard, UI development, RAG assistant integration
Yussef Ahmad	Database, Reporting & QA Engineer	Database design, reports, testing, quality assurance

Risk Assessment & Mitigation Plan:

Risk	Role	Responsibilities
Poor data quality or missing values	Reduced model accuracy	Perform data cleaning, preprocessing, and validation before training
Low prediction accuracy	Unreliable RUL estimation	Compare multiple ML models, tune hyperparameters, and evaluate using RMSE, MAE, and R^2
Difficulty interpreting AI predictions	Reduced user trust	Integrate SHAP to provide transparent feature importance explanations
Integration issues between system components	Delays in development	Adopt a modular architecture and perform continuous integration and testing
RAG retrieval returns irrelevant information	Inaccurate maintenance recommendations	Optimize text chunking, embeddings, and FAISS similarity search; improve the knowledge base
Team schedule delays	Missed project milestones	Monitor progress through weekly meetings, task tracking, and milestone reviews
Technical or software compatibility issues	Development interruptions	Use version control (Git), virtual environments, and dependency management
Limited computing resources	Longer training and processing time	Optimize code, use efficient algorithms, and leverage pre-trained models where possible

Key Performance Indicators (KPIs):

KPI Category	Metric	Target / Measurement
Model Performance	RUL Prediction Accuracy	Evaluate model performance using RMSE, MAE, and R^2 metrics to ensure reliable Remaining Useful Life estimation
Prediction Response Time	Inference Latency	Generate RUL predictions within less than 1 second after receiving valid input data
System Availability	Application Uptime	Maintain system availability above 95% during testing and demonstration phases
Dashboard Performance	Loading Time	Ensure Streamlit dashboard loads within 3 seconds under normal operating conditions
Explainability Quality	SHAP Interpretation	Provide clear feature importance explanations for model predictions
RAG Assistant Performance	Response Relevance	Provide clear feature importance explanations for model predictions
System Integration	Component Reliability	Successfully integrate ML model, XAI module, RAG assistant, backend services, and dashboard without failures
User Experience	Interface Usability	Provide an intuitive dashboard allowing users to easily monitor equipment health and access predictions
Documentation Quality	Project Completeness	Deliver complete technical documentation, reports, and final presentation materials
Team Progress	Milestone Completion	Complete all planned milestones according to the project timeline